

Ahmed Abbas

905-466-6752 | ahmedabb101@gmail.com | [linkedin.com/in/ahmedabbasc](https://www.linkedin.com/in/ahmedabbasc) | github.com/ahmedabb104

EDUCATION

University of Waterloo

Master of Data Science and Artificial Intelligence

- Coursework: Machine Learning, Statistics, Exploratory Data Analysis, Deep Learning, Data Engineering

September 2024 – April 2026

Waterloo, ON

McMaster University

Bachelor of Applied Science in Computer Science

September 2020 – April 2024

Hamilton, ON

SKILLS

Programming Languages/Databases: Python, SQL, Snowflake, MySQL, MongoDB

ML/Data Analysis: Scikit-learn, PyTorch, TensorFlow, Pandas, Matplotlib

Technologies: Apache Airflow, GCP, Docker, Power BI, Git

EXPERIENCE

Data Scientist

Thri5

May 2025 – December 2025

Toronto, ON

- Built and deployed 10+ **statistical and machine learning** data models to uncover **actionable supply chain insights** for a major retail client, uplifting profit by 3.6% and cycle count performance by 138%
- Planned and implemented a scalable **ML pipeline** in Snowflake, orchestrating with an Airflow DAG
- Maintained and deployed a **data pipeline** to support the company's data models
- Conducted a causal inference analysis on our first customer, using **statistical significance** testing (ANOVA) to validate **sales impact**, securing the startup's first commercial deal
- Analyzed large datasets** to discover and present insights to non-technical stakeholders
- Leveraged skills in **SQL** (Snowflake), Apache Airflow, exploratory data analysis, **machine learning**, and statistics

Deep Packet QA Analyst Co-op

Nokia

May 2024 – August 2024

Kanata, ON

- Designed and implemented dozens of test cases for application assurance services in Nokia's routers
- Developed a tool using **Matplotlib** to analyze and visualize network traffic stats over time, from **unstructured** text documents, aiding in identifying the root-cause of at least 3 customer-reported issues
- Generated artificial network traffic using Scapy for 12 feature tests, enhancing test coverage and robustness
- Leveraged skills in **Python**, **data visualization (Matplotlib)**, network analysis (Wireshark, Scapy) and Git

Software Engineer Intern

TD Bank Financial Group

May 2022 – August 2022

Toronto, ON

- Redesigned and developed the front-end of a KPI dashboard to improve **data visualization** and user experience
- Developed a scraper to pull and store health data from 25 monitors on Datadog, allowing the KPI dashboard to display health statuses of internal software using cron and **MongoDB**
- Onboarded the KPI dashboard onto an enterprise federation server to enable development of OIDC authentication
- Leveraged **Python**, Flask, and **MongoDB** to streamline **data processing**, and worked in an Agile environment

PROJECTS

Predicting NHL Concussions 🔄 | *Scikit-learn, Pandas, NumPy*

December 2024

- Developed supervised **machine learning** classification models (Random Forest, K-Nearest Neighbors, and XGBoost) on an imbalanced dataset to predict NHL players' next-season concussion risk using **scikit-learn**
- Conducted **data wrangling** on 10,000+ player stat records using **Pandas**, identified feature multicollinearity with Variance Inflation Factor (VIF), applied SMOTE for data augmentation, and tuned hyperparameters
- Authored a research paper comparing model performance using accuracy, precision, recall, and F1 score
- Achieved 90.1% test accuracy and 0.833 F1 score on **Random Forest** using k-fold cross-validation

Review Sentiment Analysis 🔄 | *Power BI, Transformers, Matplotlib, Pandas, NumPy*

December 2024

- Applied **NLP** sentiment analysis on 100,000 unlabeled ChatGPT reviews from Google Play using a pre-trained Hugging Face **transformer** model, revealing 74.9% positive, 11.8% negative, and 13.3% neutral reviews
- Performed **exploratory data analysis** to uncover patterns in the data through visualizations in **Matplotlib**
- Summarized findings in a business-friendly format by creating 4 visuals on a **Power BI** report